

CASABLANCA NOUASSEUR, MoroccoWMO: **601560**Lat: **33.367N**Lon: **7.590W**Elev: **200**StdP: **98.95**Time Zone: **1.00 (E01)**Period: **94-19**WBAN: **99999****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	3.0	4.2	-0.1	3.8	7.1	1.7	4.4	8.0	12.1	13.5	10.2	13.8	1.8	160	0.440

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	10.6	36.0	21.6	33.4	21.4	31.3	21.0	23.6	31.0	22.9	29.7	22.3	28.4	5.2	20

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
21.9	16.9	25.2	21.1	16.1	24.7	20.3	15.3	24.2	71.2	31.1	68.5	29.4	66.1	28.5	29.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
9.8	8.4	7.5	0.8	41.5	1.6	2.8	-0.4	43.5	-1.3	45.1	-2.2	46.7	-3.4	48.8		
			0.0	25.4	1.8	1.3	-1.3	26.3	-2.3	27.0	-3.3	27.8	-4.6	28.7		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	18.5	11.9	12.8	15.1	16.5	19.4	22.5	24.4	25.0	23.0	20.8	16.2	13.4
	DBStd	5.20	2.24	2.56	2.85	2.64	3.09	2.72	2.57	2.30	2.26	2.66	3.09	2.32
	HDD10.0	14	7	5	1	0	0	0	0	0	0	0	0	2
	HDD18.3	793	198	156	109	67	22	1	0	0	0	8	79	153
	CDD10.0	3099	67	82	158	196	293	374	446	466	391	334	187	107
	CDD18.3	836	0	1	7	13	56	124	187	207	141	83	16	1
	CDH23.3	6407	4	14	113	160	507	951	1471	1608	881	568	119	11
(14) Wind	WSAvg	3.6	3.1	3.3	3.6	3.8	4.0	4.0	4.2	3.9	3.5	3.4	3.4	3.3
	PrecAvg	360	56	42	39	32	14	2	1	0	10	36	69	59
	PrecMax	840	258	138	101	92	63	14	8	4	60	98	230	303
	PrecMin	198	2	2	1	1	2	0	0	0	0	6	3	0
	PrecStd	136	55	36	25	24	16	3	2	1	14	28	56	62
(20) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	23.2	25.3	30.2	31.9	36.9	37.9	39.9	39.0	35.8	34.1	29.9	24.9
		MCWB	15.4	15.7	17.5	17.4	19.3	21.2	22.6	22.4	21.6	19.2	17.6	14.7
	2%	DB	20.7	22.2	26.3	27.8	31.8	33.5	35.2	34.9	32.1	31.0	26.5	22.2
		MCWB	14.5	15.3	16.5	17.9	19.5	21.1	22.0	22.6	21.7	19.0	17.2	14.8
	5%	DB	18.8	20.0	23.4	24.6	28.2	30.8	32.1	32.1	29.5	28.7	24.1	20.1
		MCWB	14.2	14.9	16.3	17.4	18.5	20.7	21.7	22.1	21.2	19.3	17.0	14.9
	10%	DB	17.1	18.1	21.0	22.1	25.6	28.1	29.9	30.0	27.8	26.3	22.0	18.6
		MCWB	13.6	14.1	15.6	16.3	18.1	20.2	21.4	21.8	20.7	19.3	16.7	14.6
	0.4%	WB	17.1	17.9	19.3	20.5	21.7	23.2	24.7	24.6	23.7	22.1	20.5	17.8
		MCDB	20.6	22.7	25.8	27.6	31.6	32.7	33.9	32.5	30.2	28.3	24.8	20.8
(30) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	15.7	16.3	17.8	18.6	20.3	22.0	23.3	23.7	22.7	21.1	19.1	16.6
		MCDB	18.8	20.1	23.6	25.0	28.6	30.0	31.3	30.9	28.3	26.4	23.1	19.4
	5%	WB	14.7	15.3	16.8	17.6	19.4	21.2	22.6	23.0	22.0	20.4	18.0	15.8
		MCDB	17.6	18.9	22.1	23.2	26.8	28.6	29.7	29.6	26.9	25.2	21.9	18.5
	10%	WB	13.9	14.5	15.9	16.7	18.5	20.5	21.8	22.4	21.3	19.8	17.1	15.0
		MCDB	16.7	17.7	20.3	21.6	24.5	26.9	28.1	28.2	26.0	24.1	21.0	17.8
(36) Mean Daily Temperature Range	5% DB	MDBR	11.0	11.3	11.8	11.2	11.3	10.8	10.5	10.6	10.5	11.5	11.1	10.8
		MCDBR	13.6	13.9	16.1	15.6	16.5	15.4	14.8	13.8	13.2	15.1	14.0	13.6
		MCWBR	8.9	8.6	8.7	7.7	6.7	5.6	5.0	4.6	5.0	6.7	7.5	8.3
	5% WB	MCDBR	11.5	12.2	13.6	13.2	13.8	12.8	12.7	12.1	11.3	12.0	11.2	10.3
		MCWBR	8.1	8.0	8.0	7.1	6.1	5.2	4.9	4.5	4.7	6.0	6.5	7.0
(40) Clear-Sky Solar Irradiance	taub		0.326	0.354	0.407	0.404	0.429	0.462	0.506	0.500	0.441	0.401	0.361	0.335
	taud		2.493	2.398	2.224	2.264	2.208	2.114	1.976	2.014	2.211	2.340	2.441	2.496
	Ebn at Noon		889	894	866	885	862	830	791	789	823	830	838	856
	Edh at Noon		86	105	135	135	145	159	181	172	135	110	91	82
(44) All-Sky Solar Radiation	RadAvg		3.06	3.91	5.06	6.17	6.93	7.53	7.56	6.84	5.63	4.23	3.22	2.80
	RadStd		0.20	0.34	0.34	0.30	0.38	0.30	0.16	0.19	0.23	0.27	0.22	0.18

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	+0.91	N/A	N/A	N/A	N/A	N/A	+109
(47) Regional (4 neighbors)	N/A	N/A	-0.86	+0.93	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page